> laws at the [instrumental] or macro-level which are sacrosanct, and which the micro-theories explain. We must insist… that the micro-theories can directly explain observations, such as the outcome of the two-slit experiment.

Smart and Putnam were clear-eyed about the challenges faced by any alternative to the Copenhagen interpretation. “Any realistic philosophy of the theoretical entities must not be too naïve. It must reckon with the very real difficulties in giving a non-[instrumentalist] interpretation of physics,” wrote Smart. “One way out of the dilemma may lie in the development of deterministic theories of microphysics on the lines foreshadowed by such writers as D. Bohm and J.-P. Vigier.” Putnam agreed that “something is wrong with the [quantum] theory.” But he thought von Neumann’s proof ruled out Bohm’s pilot-wave interpretation—at the time, Bell’s refutation of the proof was still languishing in an editor’s desk—and he was wholly unaware of Everett’s many-worlds interpretation (as were Smart and almost everyone else). Putnam concluded that “*no* satisfactory interpretation of quantum mechanics exists today.” But he was hopeful that the problem would be solved. “Human curiosity will not rest until those questions [of quantum interpretation] are answered.… The first step toward answering them has been attempted here. It is the modest but essential step of becoming clear on the nature and magnitude of the difficulties.”

Yet this is precisely what physicists, as a whole, were still unclear about. The philosophers had successfully overthrown positivism and had a good understanding of the mathematical intricacies of quantum physics—but the physicists were still blinkered, walled off from philosophy and the developments there. They had no idea any of this had happened. While Einstein and Bohr’s generation was widely schooled in philosophy, the push toward specialization after World War II had taken its toll on the liberal arts education of the new crop of physicists. Academic departments had become Balkanized as they had grown in the postwar boom, and physicists, busy with enormous grants and hard-nosed calculations, were generally dismissive of philosophy. So physics trudged along, not knowing that a major revolution had happened in an adjacent field. And philosophers were, mostly, not surprised by this. “Unless the real difficulties in quantum mechanics can be dealt with,” wrote Smart, “the philosophical objections to the Copenhagen interpretation, which consist only in exposing the